

PES1UG24CS401 - Saiesh Shankar

SE LAB - 4 (Vibe Coding)

Prompting evidence link :

<https://chatgpt.com/share/6ac73b14-1f04-83e8-8495-10a882b4774d>

(NOTE : So the 5th prompt is completely unrelated, I accidentally posted an error from my ML Miniproject to it without realizing that this was the chat I had to submit - so please ignore that. Otherwise the tasks have been completed within four prompts)

Before and After videos :

<https://drive.google.com/drive/folders/10pFHmYX07d9Z-FWt6uAZHz2MPzobsrvE?usp=sharing>

Some Valuable code snippets :

(Task - 1)

```
def bezier(p0, p1, p2, p3, t):
```

```
    u = 1 - t
```

```
    x = (
```

```
        u**3 * p0[0]
```

```
        + 3 * u**2 * t * p1[0]
```

```
        + 3 * u * t**2 * p2[0]
```

```
        + t**3 * p3[0]
```

```
)
```

```
    y = (
```

```
        u**3 * p0[1]
```

```
        + 3 * u**2 * t * p1[1]
```

```
        + 3 * u * t**2 * p2[1]
```

```
        + t**3 * p3[1]
```

```
)
```

```
    return pygame.Vector2(x, y)
```

Fixed bezier function

(Task - 2)

```
def enemy_tint(kind):
    if kind == "boss":
        return None

    wave = max(1, WAVE_NUMBER)

    if kind == "red":
        palette = [
            (255, 60, 60),
            (255, 100, 40),
            (255, 60, 140),
            (220, 50, 255),
            (255, 150, 50),
        ]
    elif kind == "blue":
        palette = [
            (60, 140, 255),
            (40, 200, 255),
            (80, 100, 255),
            (120, 70, 255),
            (50, 220, 200),
        ]
    else:
        return None

    return palette[(wave - 1) % len(palette)]
```

Task - 3

```
if WAVE_BANNER > 0:
    banner = font.render(f"WAVE {self.wave}", True, (255, 255, 120))
    screen.blit(
        banner,
        banner.get_rect(center=(WIDTH // 2, HEIGHT // 2 - 80))
    )
Inserted at the end of Game.draw
```

Task - 4

```
def shield_charges(wave):
```

```
return 1 if wave % 3 == 0 else 0
```